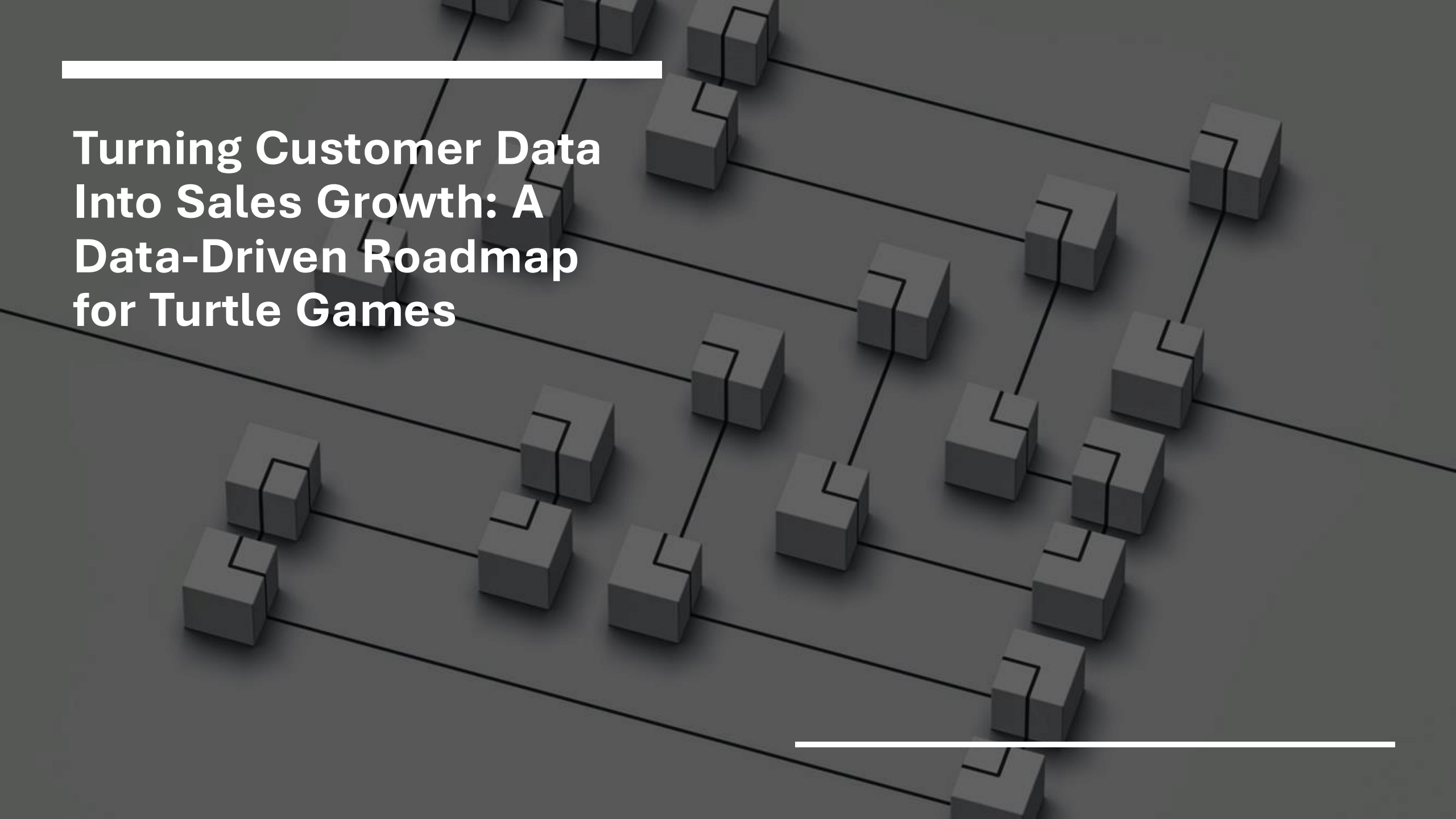
Turning Customer Data Into Sales Growth: A Data-Driven Roadmap for Turtle Games



The Business Challenge

Turtle Games is a manufacturer and retailer of books, board games, video games, and toys to a global customer base.

Objective: improve sales performance through customer trend analysis.

Four guiding questions: how loyalty points are earned, how to segment customers, how reviews can inform marketing, and if the data available supports reliable prediction.

Analytical Approach

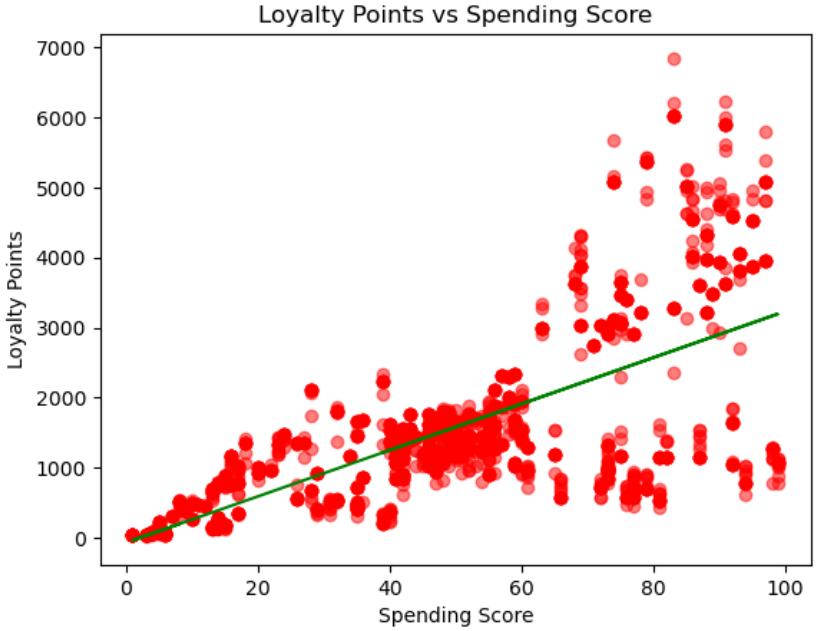
Tested simple relationships first, then moved to more advanced models, only adding complexity where the data justified it.

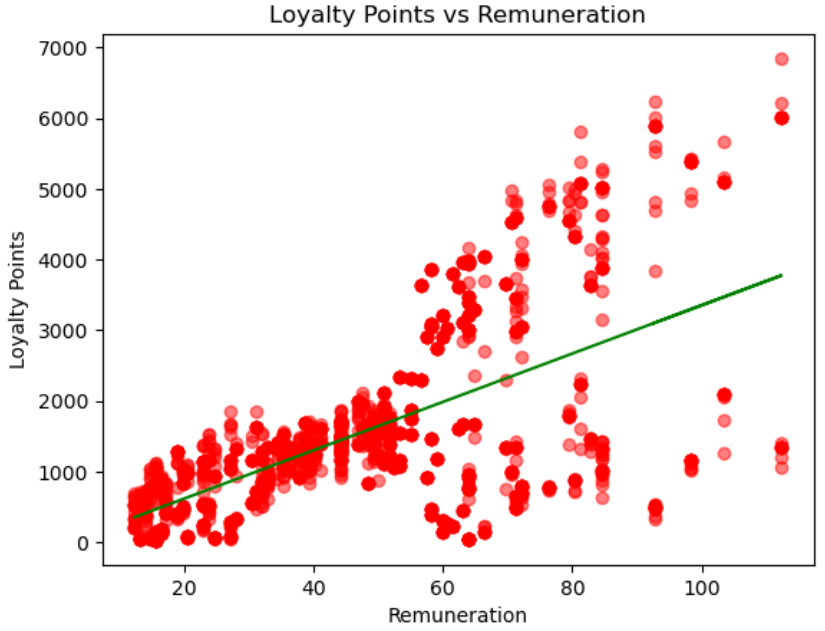
Used two independent methods to validate each major finding before trusting it.

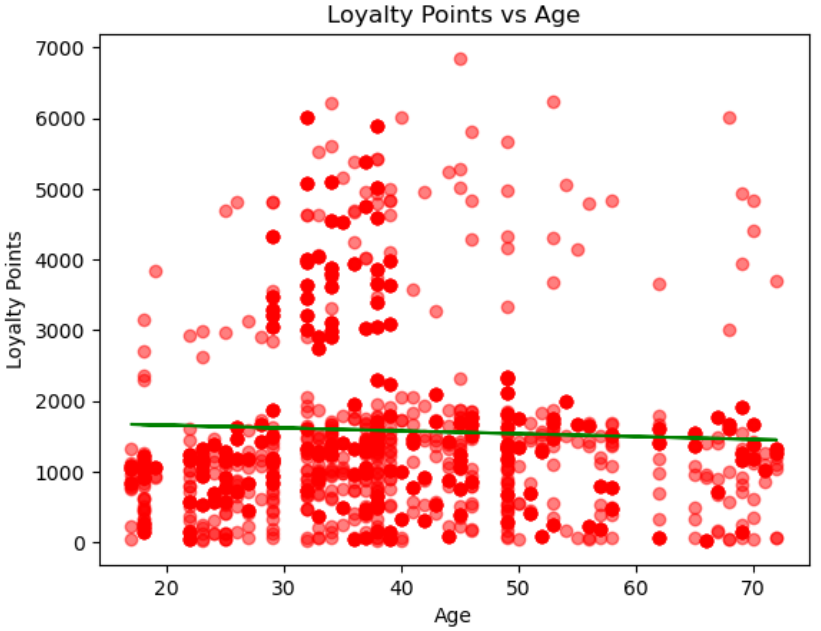
Cross-checked results in two different analytical tools to confirm nothing was a fluke of one platform.

Insight 1: What Actually Drives Loyalty Points

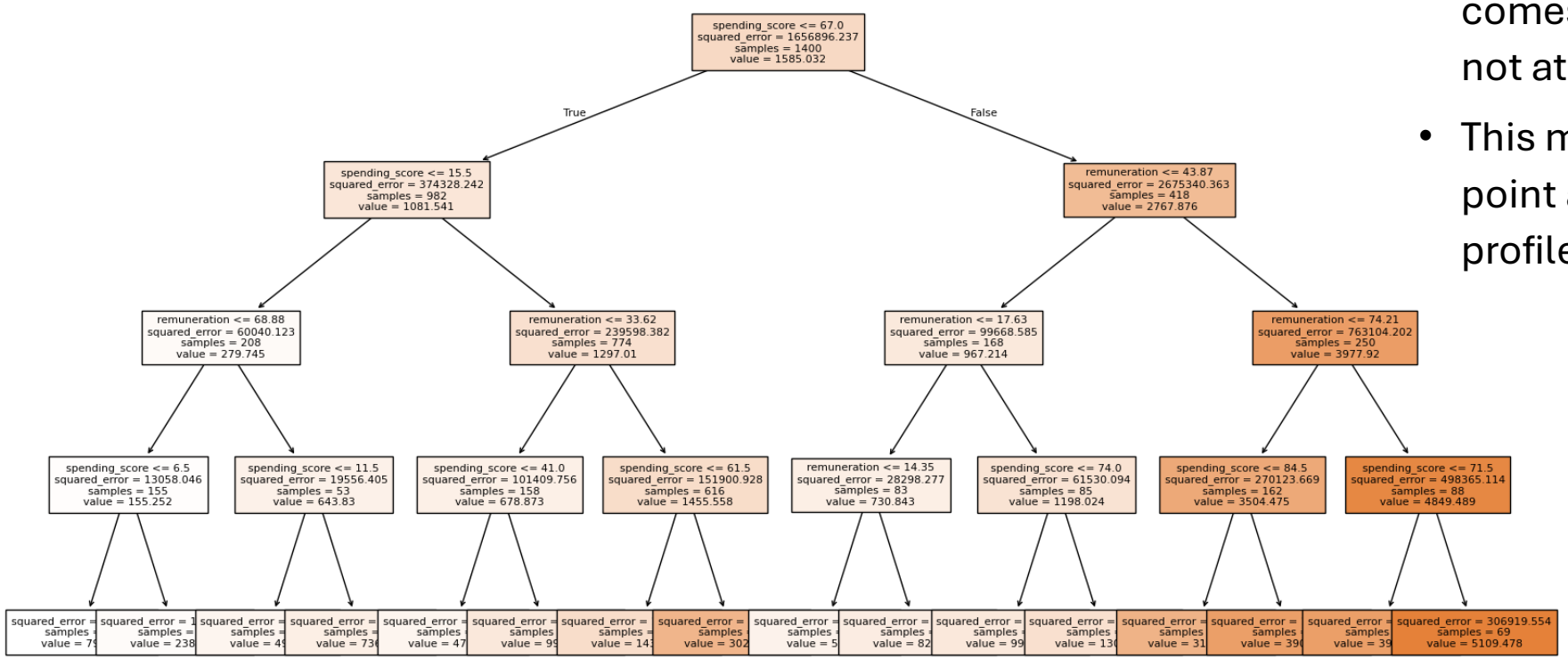
- Tested three possible drivers of loyalty points: spending behaviour, income, and age.
- Spending score is the strongest single driver, but remuneration also plays an independent role. Both reveal distinct customer bands.
- Age essentially has no relationship as a driver of loyalty points. Age-based marketing assumptions about loyalty will not work for this customer base.







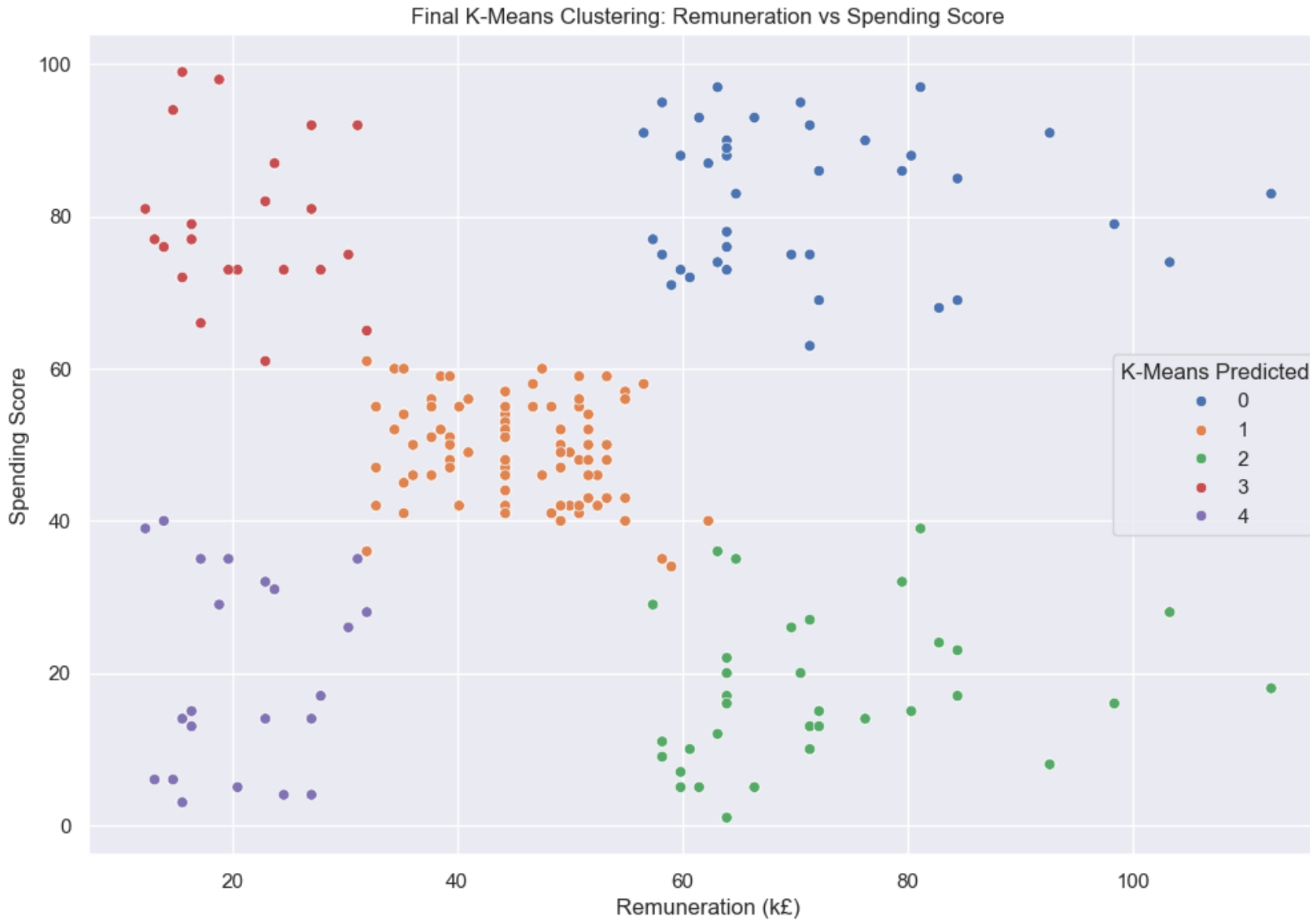
Insight 2: Predicting Loyalty Points With High Confidence



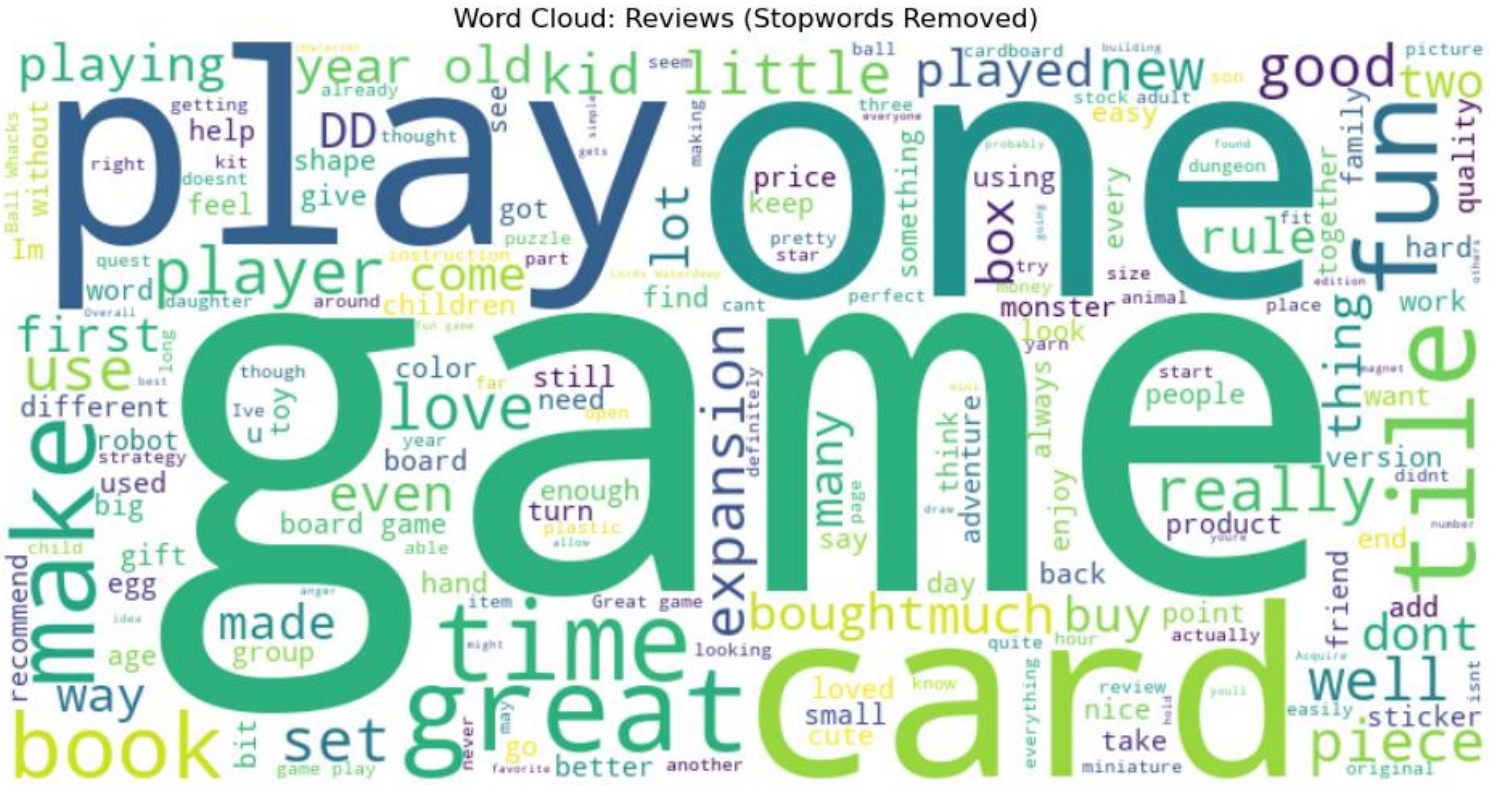
- A predictive model built that explains 94% of why customers earn the loyalty points they do.
- The model confirms: spending comes first, income second, and age not at all.
- This model can now forecast loyalty point accumulation for any customer profile.

Insight 3: Five Distinct Customer Segments Exist

- Statistically validated five customer segments, each with distinct income and spending behaviour.
- One standout opportunity: 330 customers (16.5% of the customer database) have high income but low spend.
- This group alone presents a clear, targetable path to increasing revenue.



Insight 4: What Customers Are Actually Saying



Negative Reviews	review
BOOO UNLES YOU ARE PATIENT KNOW HOW TO MEASUR...	
Incomplete kit Very disappointing	
One of my staff will be using this game soon s...	
I bought this as a Christmas gift for my grand...	
I sent this product to my granddaughter The po...	
My 8 yearold granddaughter and I were very fru...	
I purchased this on the recommendation of two ...	
If you like me used to play DD but now you and...	
My son loves playing this game It was recommen...	

Positive Reviews	review
Perfect	
My daughter loves her stickers Awesome seller ...	
Perfect for tutoring my grandson in spelling	
The best part I see is the box What a wonderf...	
Great quality very cute and perfect for my to...	
This is a great tool to have at hand when play...	
Great price arrived on time with no damage Wil...	
The pictures are great Ive done one and gave ...	
Prompt service and a great product	

Recommendations

1. Redesign the loyalty programme into spending-based tiers.
2. Launch a targeted campaign for the high-income, low-spend segment.
3. Route recurring review complaints to product quality control.
4. Lean into gift-giving and family-learning messaging.

Conclusion & Next Steps

- Loyalty points are earned through spending behaviour, not demographics.
- Five real customer segments exist, with one clear, quantified opportunity.
- Reviews reveal fixable product issues and an untapped marketing angle.
- Prediction with the available data can be done with high confidence.

Further Research

- Expand review analysis beyond the business website, e.g. social and video platform comments.